

1. Fig. 1 (a) is an illustration of a system found in humans and Fig.1 (b) shows the illustration of the structure of a functional unit of the part labelled IV.

Study the figure carefully and answer the questions that follow.

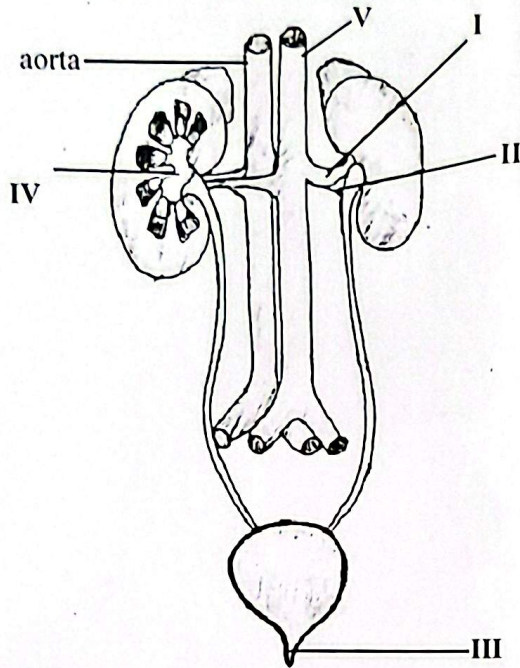


Fig. 1(a)

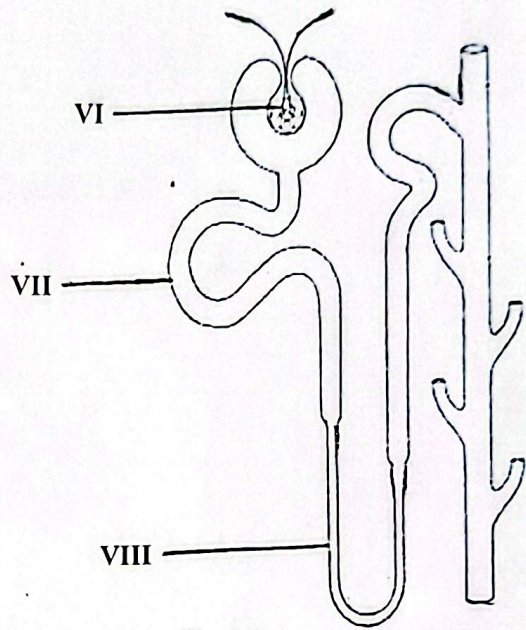


Fig. 1(b)

- (a) Name the:
- system illustrated in Fig. 1(a);
 - functional unit illustrated in Fig. 1(b).

[2 marks]

- (b) Name each of the parts labelled V, VI, VII and VIII.

V.....
 VI.....
 VII.....
 VIII.....

[4 marks]

- (c) Explain the function of each of the parts labelled I, II, III, VI and VIII.

I.....

 II.....

 III.....

VI.....

.....

VIII.....

.....

(d) State two observable features of the part labelled:

[10 marks]

(i) VI, which enhance its function;

.....

.....

(ii) VIII, which enhance its function.

.....

.....

[4 marks]

2. Fig. 2 is a model of pH-scale indicating the acidic and basic strengths of some selected chemicals used at home. The selected chemicals are potash solution, lemon juice, soap solution, table salt solution, hydrogen peroxide, hydrochloric acid and washing soda solution. These chemicals are represented as A, B, C, D, E, F and G, but not in any particular order.

Study the figure carefully and answer the questions that follow.

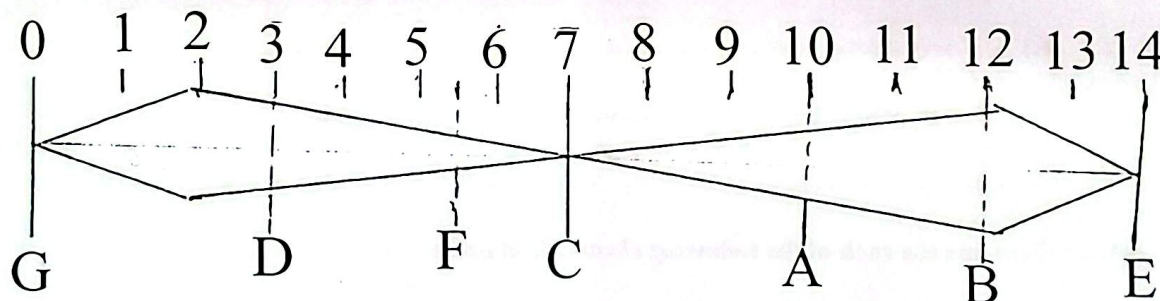


Fig. 2

(a) Fill in the blank spaces in the table provided by assigning:

- (i) each of the named chemicals to the corresponding correct label;
- (ii) the pH-value on the pH-scale;
- (iii) the colour of the chemical upon addition of a universal indicator.

Label of Chemical	pH of chemical	Colour of Universal Indicator	Name of Chemical at home
A	10		Soap
B	12	Blue	
C	7		xxx
D			xxx
E	14	xxx	
F	xxx	Lime green	Hydrogen peroxide
G	0		

[8 marks]

- (b) (i) Select **two** chemicals listed which can react to give a product with $\text{pH} = 7$.

.....
 [2 marks]

- (ii) Name the reaction in (i).

..... [1 mark]

- (c) Identify the chemical that **cannot** change the colour of red litmus and blue litmus.

..... [1 mark]

- (d) State the colour of methyl orange in:

(i) **D**;

(ii) **E**.....

[2 marks]

- (e) State the observation that would be made when **B** is added to:

(i) **G**;.....

(ii) phenolphthalein

[2 marks]

- (f) Identify the solution that could be classified as:

(i) very strong acid;.....

(ii) very strong base.....

[2 marks]

- (g) Give **one** use **each** of the following chemicals at home:

(i) **B**;.....

(ii) **C**.....

[2 marks]

?

3. In an experiment to investigate the effect of various solutions on potato cells, slices of potato were placed in equal volumes of **tapwater**, **distilled water**, and **salt water** for 48 hours. The slices were carefully removed from the solutions and their sizes measured again and compared.

(a) State **two** observations that would be made on the slices from **each** solution provided:

Tapwater;.....

.....

Distilled water;.....

.....

Salt water.....

.....

[6 marks]

(b) Explain the observations made in (a).

.....

.....

.....

.....

.....

.....

.....

.....

[8 marks]

(c) State the conclusion that could be drawn from the results obtained in the experiment.

.....

.....

.....

.....

[2 marks]

- (d) Sketch and label a diagram to show the appearance of the cell placed in the:
- (i) distilled water;

- (ii) salt water.

[4 marks]

4. A learner wanted to use a piece of glass as a lens for the eyes. The learner was advised to determine the refractive index of the glass and was given optical pins and a protractor. The learner was directed to use any five angles of incidence and to find their corresponding angles of refraction. Fig. 4 shows angles of incidence $i_1^\circ, i_2^\circ, i_3^\circ, i_4^\circ$, and i_5° and their corresponding angles $r_1^\circ, r_2^\circ, r_3^\circ, r_4^\circ$, and r_5° . Study the Figure carefully and answer the questions that follow.

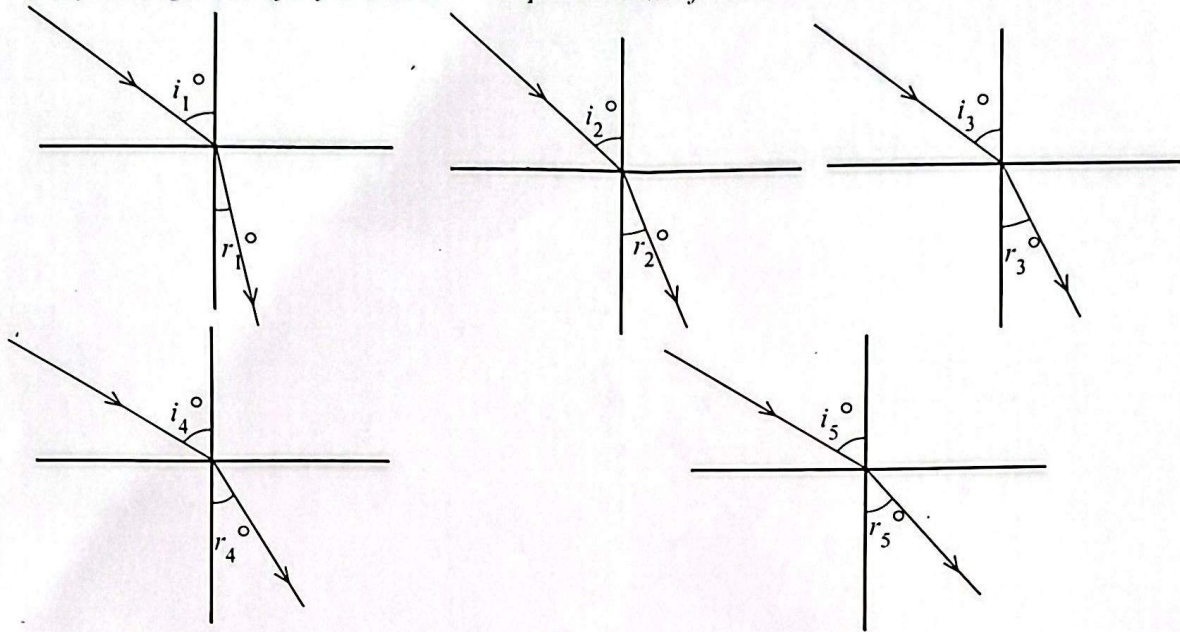


Fig. 4

- Measure and record the angles of incidence, $i^\circ = i_1^\circ, i_2^\circ, i_3^\circ, i_4^\circ$, and i_5° .
- Measure and record the corresponding angles of refraction, $r^\circ = r_1^\circ, r_2^\circ, r_3^\circ, r_4^\circ$, and r_5° .
- Determine the sine of the angles of incidence and their corresponding angles of refraction.
- Record the values obtained in the table provided.

Angle of incidence $i / ^\circ$	$i_1 =$	$i_2 =$	$i_3 =$	$i_4 =$	$i_5 =$
Angle of refraction $r / ^\circ$	$r_1 =$	$r_2 =$	$r_3 =$	$r_4 =$	$r_5 =$
$\sin i$	$\sin i_1 =$	$\sin i_2 =$	$\sin i_3 =$	$\sin i_4 =$	$\sin i_5 =$
$\sin r$	$\sin r_1 =$	$\sin r_2 =$	$\sin r_3 =$	$\sin r_4 =$	$\sin r_5 =$

- Plot a graph of $\sin i$ on the vertical axis and $\sin r$ on the horizontal axis.

[10 marks]

- Determine the slope of the graph.

[5 marks]

- State the significance of the slope.

[3 marks]

[2 marks]

END OF PAPER

